

EXPERIMENT 1

Aim

To study and implement Linear Regression and Logistic Regression models on a real-world student performance dataset for predicting final exam scores and classifying students as pass or fail.

Problem Statement

Student performance depends on multiple factors such as study time, attendance, and internal assessment scores. The problem is to develop machine learning models that can:

- Predict the **final exam score (continuous value)** using Linear Regression
- Classify whether a student will **pass or fail (binary outcome)** using Logistic Regression

The goal is to analyze how different academic factors influence performance and build accurate predictive models.

Dataset Link

<https://www.kaggle.com/code/mohamedredaibrahim/student-performance-dataset>

Methodology

1. **Data Collection**
 - Loaded the student performance dataset from CSV file
2. **Data Preprocessing**
 - Removed missing/null values
 - Selected relevant features:
 - Hours_Studied
 - Attendance
 - Assignment_Score
 - Midterm_Score
3. **Feature & Target Separation**
 - Input Features (X): Study and performance attributes
 - Target Variable (y): Final_Score
4. **Data Splitting**
 - Split dataset into training and testing sets (80:20 ratio)

5. Linear Regression Model

- Applied Linear Regression to predict continuous final scores
- Model learns relationship between features and final marks

6. Logistic Regression Model

- Converted Final_Score into binary:
 - Pass (≥ 65) $\rightarrow 1$
 - Fail (< 65) $\rightarrow 0$
- Applied Logistic Regression for classification

7. Model Evaluation

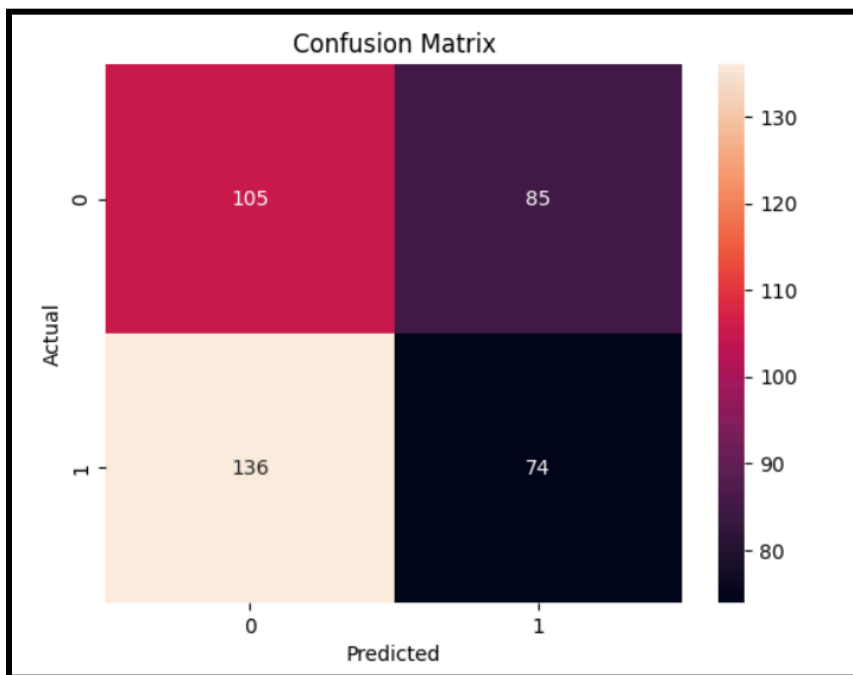
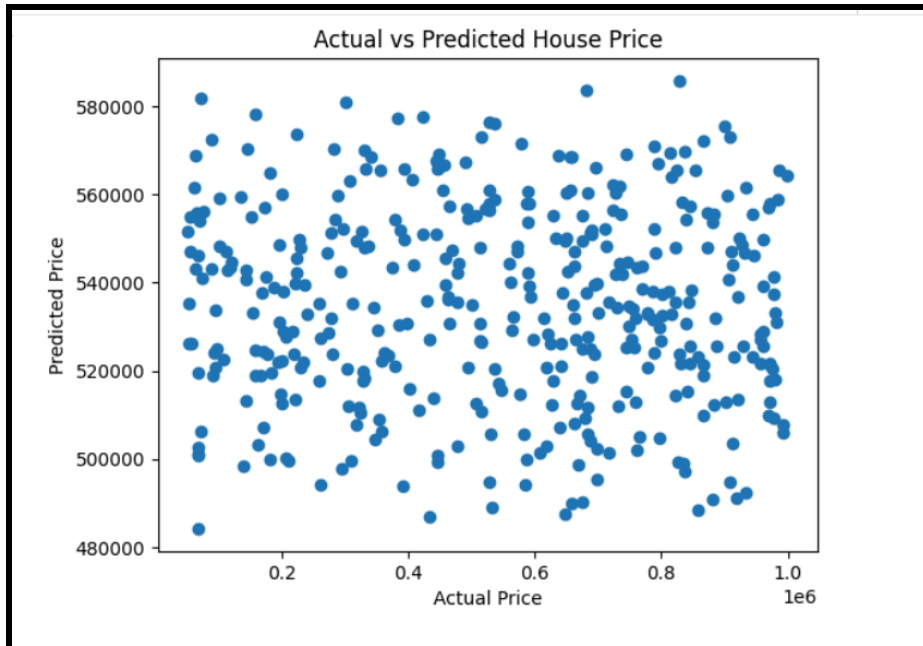
- Linear Regression:
 - Mean Absolute Error (MAE)
 - Mean Squared Error (MSE)
 - R^2 Score
- Logistic Regression:
 - Accuracy
 - Confusion Matrix
 - Precision, Recall, F1-score

8. Hyperparameter Tuning

- Used GridSearchCV for Logistic Regression
- Tested different values of parameter C and solvers

Screenshots :

```
===== LINEAR REGRESSION RESULTS =====  
MAE: 244419.99901363705  
MSE: 78729422262.64821  
RMSE: 280587.637401665  
R2 Score: -0.011961539273188437
```

**Conclusion:**

This experiment demonstrated the application of Linear and Logistic Regression techniques for prediction and classification tasks. Linear Regression effectively predicted continuous output values, while Logistic Regression accurately classified data into binary categories. Model performance was evaluated using appropriate metrics, and hyperparameter tuning improved classification accuracy. The experiment highlights the importance of regression models in supervised machine learning and real-world data analysis.